

Practical 9: Cleaning and Structuring Data for Analysis

AIM:a) Identify and handle missing or inconsistent data

b) Convert wide data into tall format

c) Perform union of multiple files

d) Execute cross-database joins

e) Restructure data for better visualization

f) Handle different levels of detail in datasets

g) Apply advanced data cleaning techniques

a) Identify and handle missing or inconsistent data

Step 1: Load the Dataset

Tableau - TYCS_T110_PRC_9

File Data Server Window Help

Connections

Sem1_Scores

Sheets

Use Data Interpreter
Data Interpreter might be able to clean your Microsoft Excel workbook.

Department_Master

Sem1_Scores

Sem2_Scores

Student_Activities

New Union

New Table Extension

Sem1_Scores (Sem1_Scores)

Connection: Live

Filters: Add

Sem1_Scores

8 fields 10 rows

10 rows

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Att
S101	Aarav Sharma	CSE-01	85	90	88	95	
S102	Priya Verma	ECE-02	null	75	80	85	
S103	Rohan Gupta	CSE-01	78	82	null	88	
S104	Sneha Patel	IT-03	92	88	90	96	
S105	Aditya Rao	MECH-04	45	null	50	60	
S106	Ananya Singh	CSE-01	88	91	85	92	

Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric MEASURE)

Tableau - TYCS_T110_PRC_9

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Show Me

Data

Sem1_Scores

Sem1_Scores (Sem1_Sc...

Tables

Dept Info

Full Name

Student ID

Measure Names

Attendance

Attendance_Clean

Chemistry

Chemistry_clean

Math

Math_Clean

Physics

Physics_Clean

Programming

Sem1_Scores (Count)

Measure Values

Filters

Marks

Automatic

Color

Size

Detail

Tooltip

Drop field here

Drop field here

Drop field here

Chemistry_clean

Sem1_Scores (Sem1_Scores)

IFNULL([Chemistry], 0)

The calculation is valid.

Apply

OK

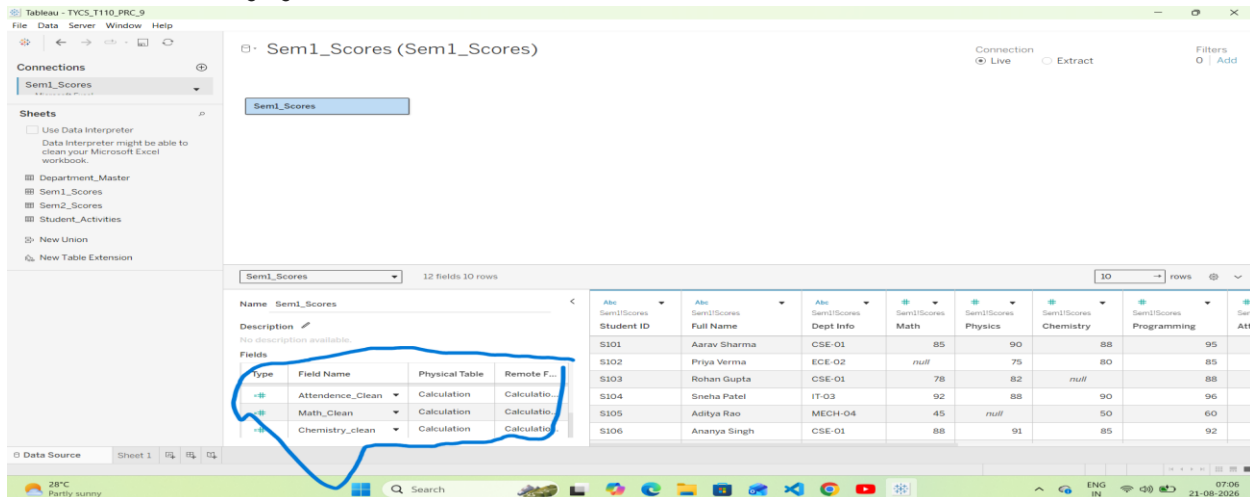
Add data to visualize
Double-click or drag fields from the data pane.

B: Convert Wide Data into Tall Format

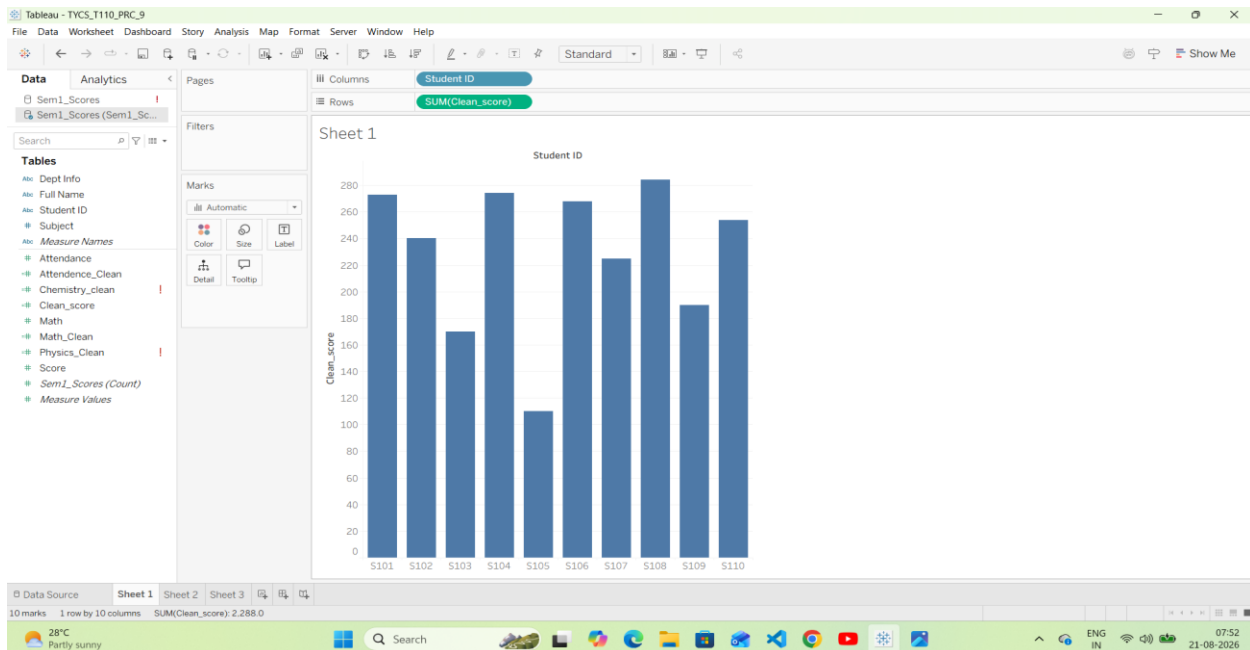
1. In Tableau, navigate to the Data Source tab at the bottom-left corner.
2. Locate the four horizontal subject score columns: o Math

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.



3. Right-click on any of the highlighted column headers.
4. Select Pivot



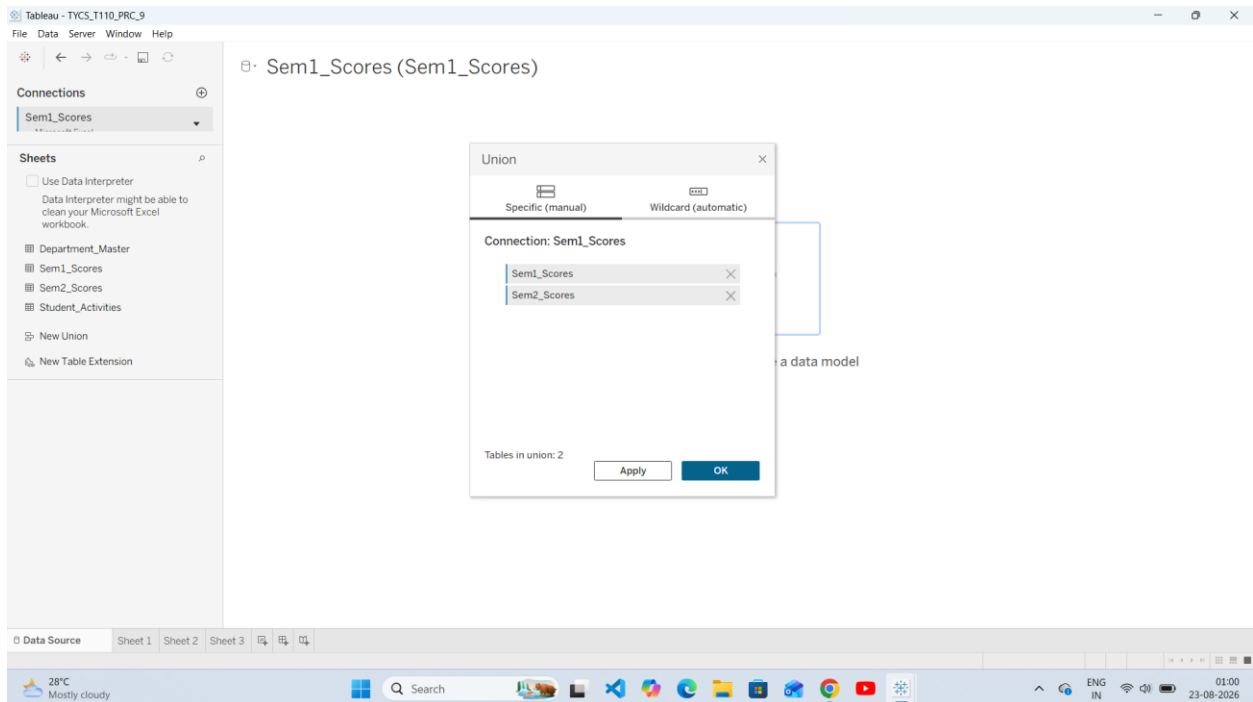
C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start

with a clean canvas.

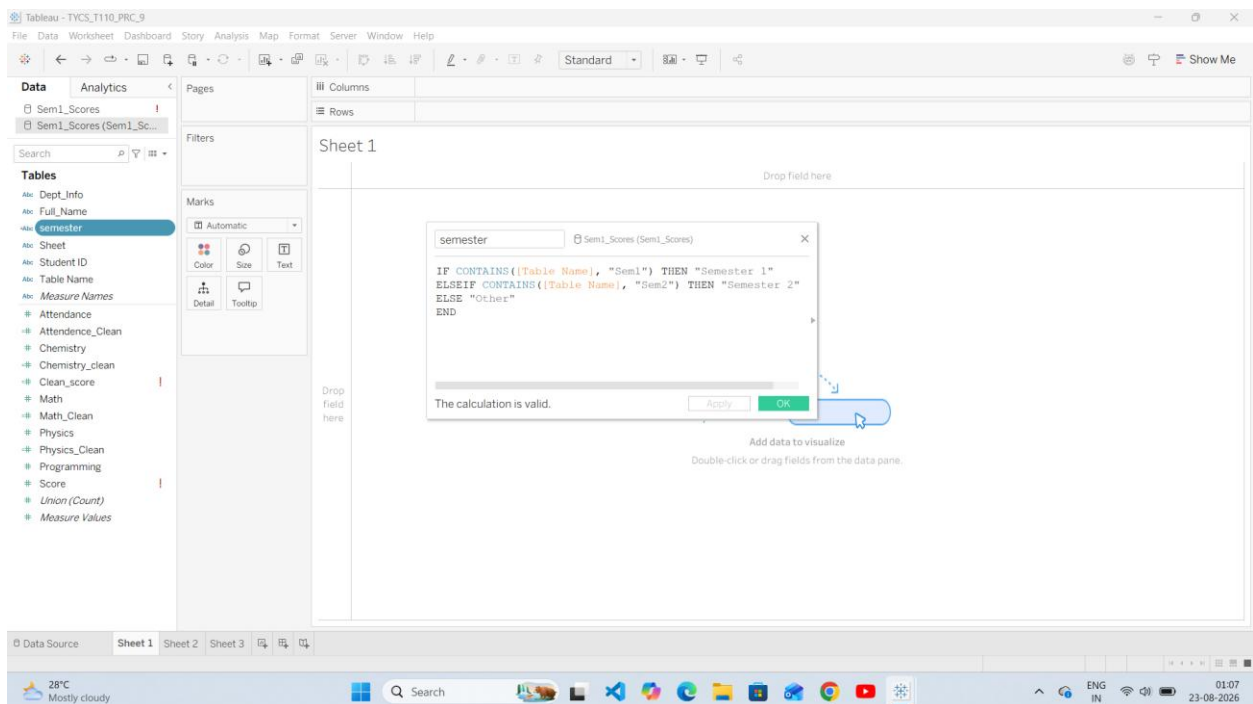
Step 2: Create a Multi-File Union



Step 3: Verify the Merged Data Grid

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
3. Set the Field Name to: Semester
4. Enter the following formula:



D: Execute Cross-Database Joins

Step 1: Stay on the Top Canvas Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

The screenshot displays the Tableau Desktop interface. On the left sidebar, the 'Connections' pane shows 'Sem1_Scores' selected. The 'Sheets' pane lists 'Department_Master', 'Sem1_Scores', 'Sem2_Scores', 'Student_Activities', 'New Union', and 'New Table Extension'. The main view area shows a Union relationship between 'Sem1_Scores' and 'Department_Master'. Below this, a data table is displayed with the following columns: Department/Master, Dept Code, Department Name, HOD Name, Dept Target Average, and Total Dept Capacity. The data table contains four rows of data.

Department/Master	Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Raman	85	120	
ECE-02	Electronics & Comm	Dr. Iyer	78	90	
IT-03	Information Tech	Dr. Sen	82	60	
MECH-04	Mechanical Engg	Dr. Sharma	70	60	

Step 3: Add and Link Student_Activities.csv

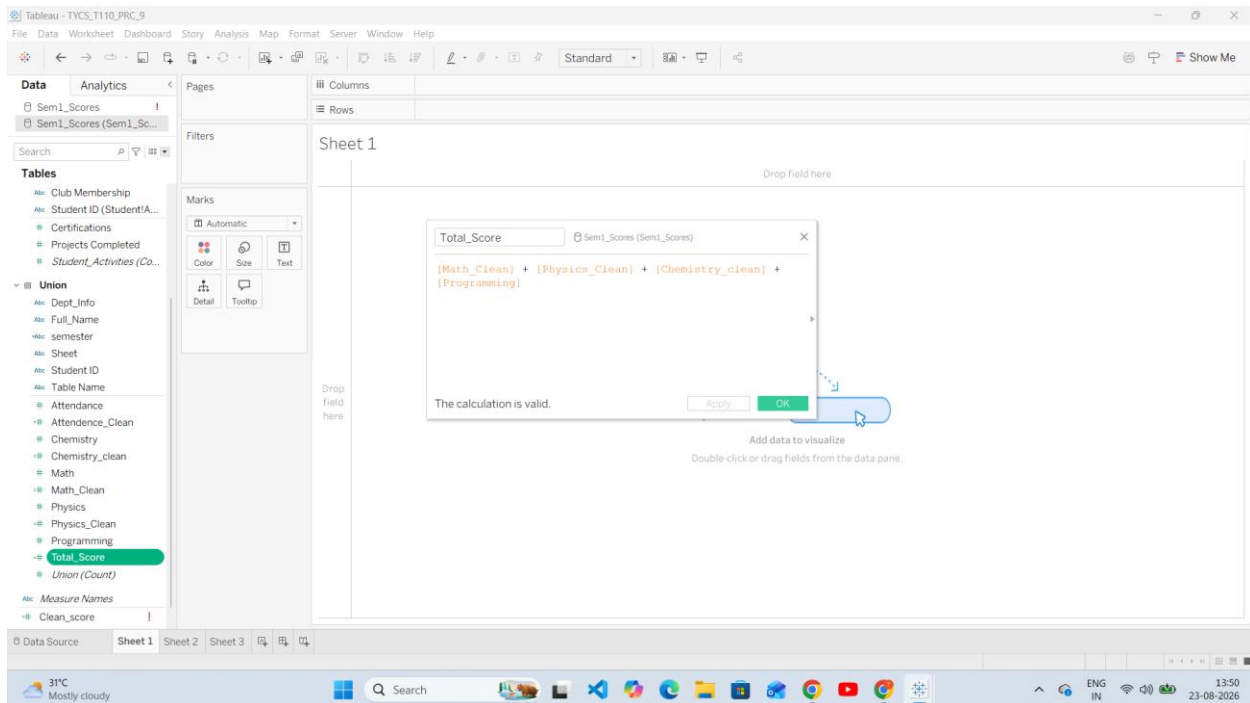
1. From the left sidebar, drag `Student_Activities.csv` onto the canvas.
2. Drop it to the right of the Union box.

The screenshot shows the Tableau Desktop interface. The top menu bar includes File, Data, Server, Window, and Help. The Connections pane on the left shows 'Sem1_Scores' selected. The Sheets pane on the left shows 'Department_Master', 'Sem1_Scores', 'Sem2_Scores', and 'Student_Activities'. The main view displays a Union of 'Department_Master' and 'Student_Activities' tables. The data table below shows the following columns and rows:

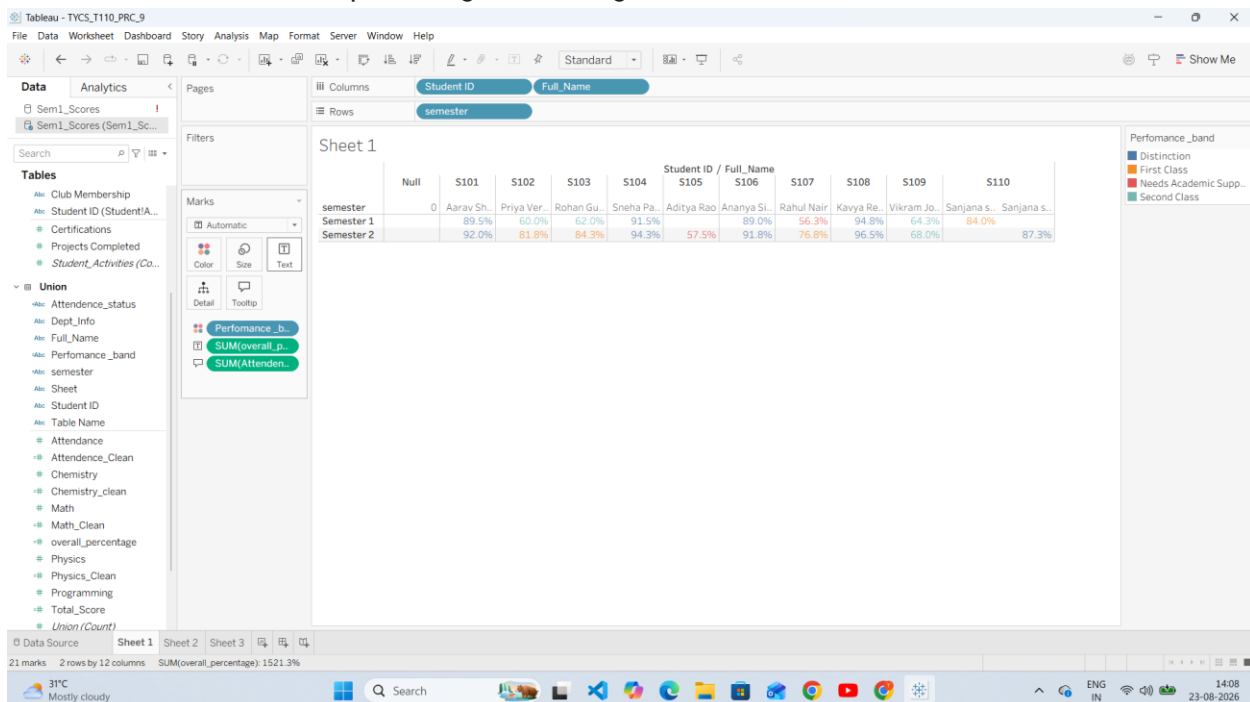
Student ID (Student)Activ	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2

E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage

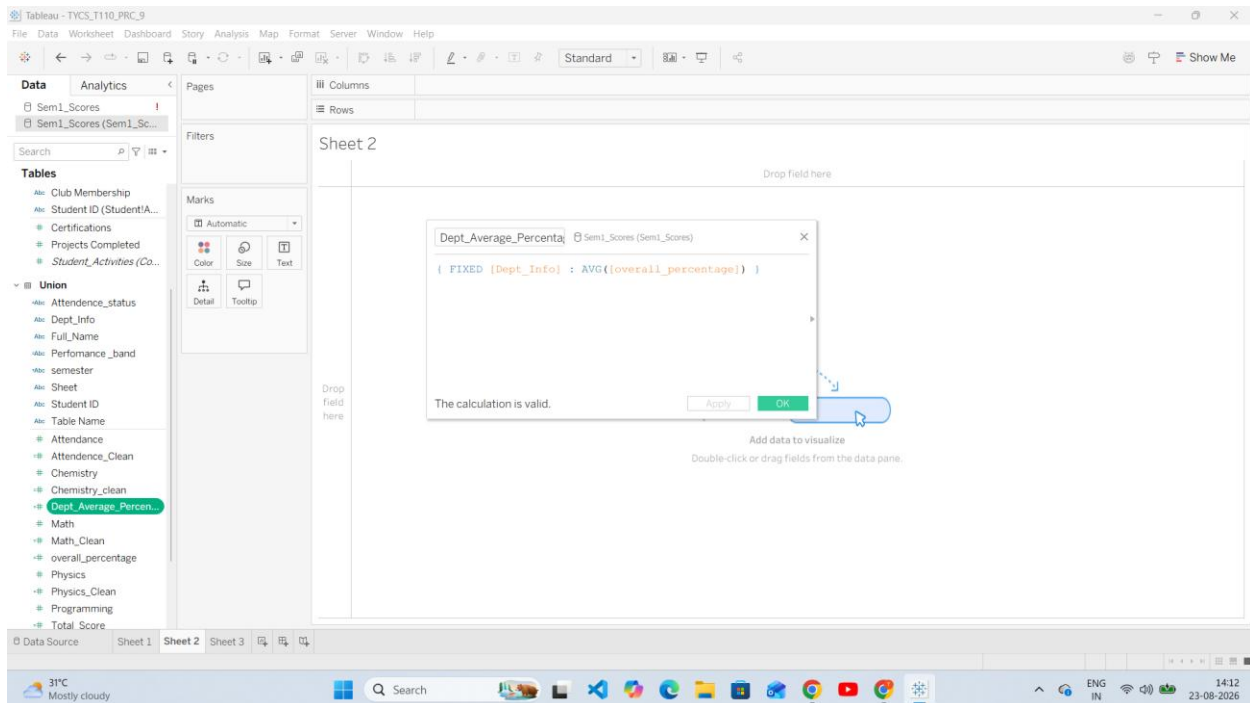


4. Create the scaled overall percentage field: o Right-click in the Data Pane --> select Create



F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD



4. Set default formatting: o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.

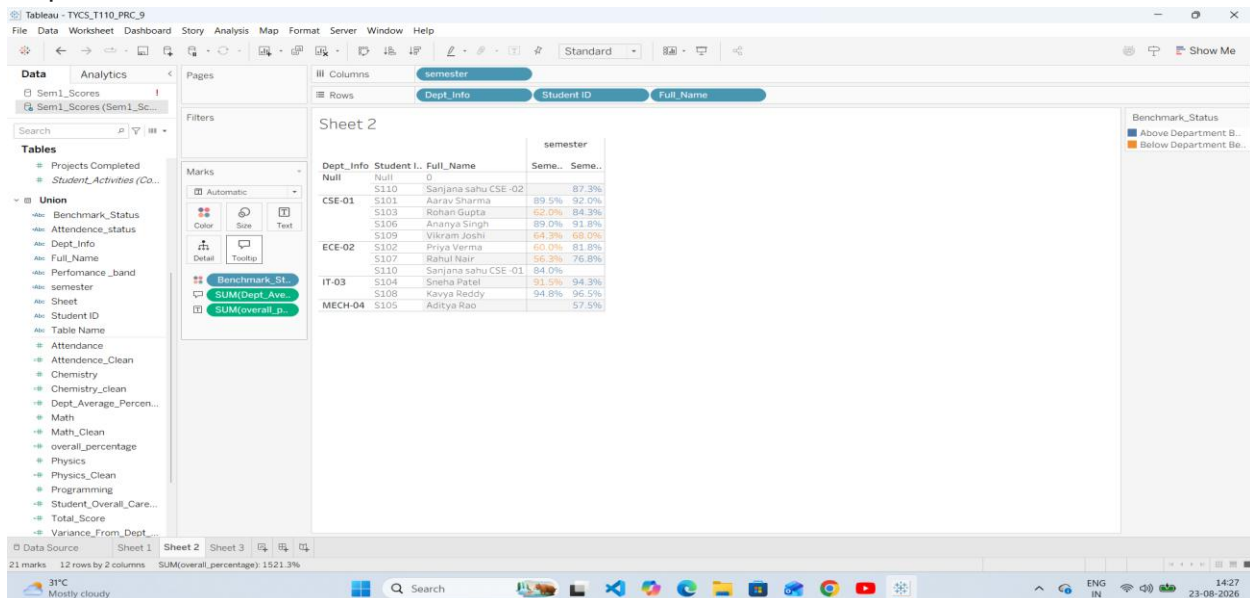
Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous

3. Set its Number Format to Percentage (1 decimal place).

Step 4: Create a Benchmark Indicator Dimension

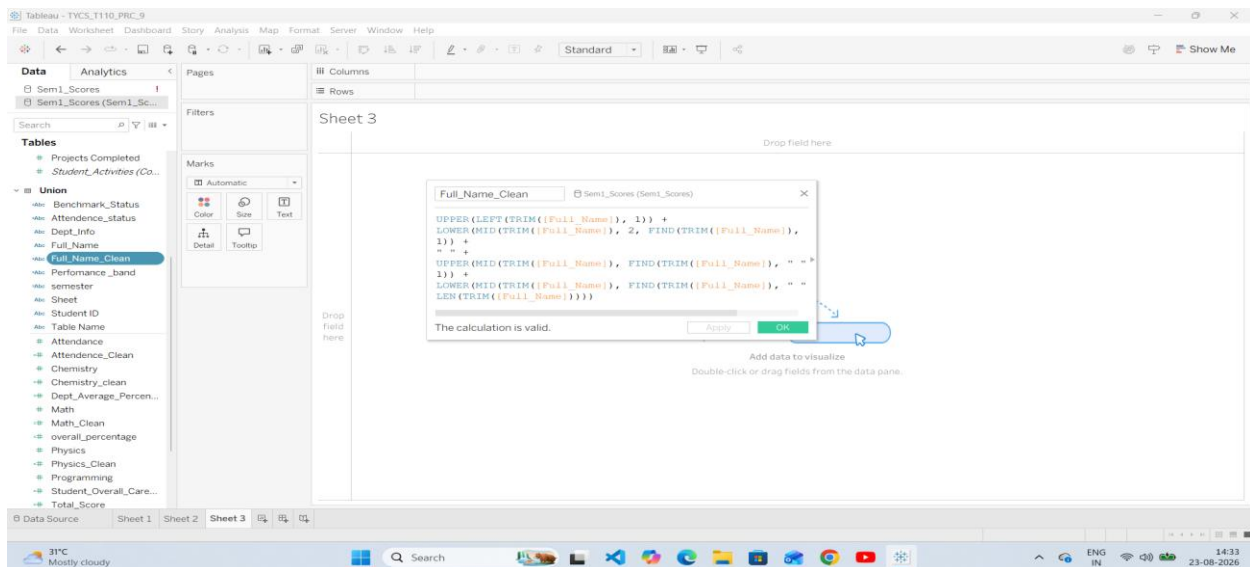
Step 5: Build the LOD Benchmark Visualization



G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

TYCS T110 SANJANA SAHU



Step 2: Parse First Name and Last Name using SPLIT().

2. Extract the Last Name: o Right-click in the Data Pane --> select Create Calculated Field.... o Field

Step 3: Split Composite Department Codes (Dept_Info):

Step 4: Create a Clean Composite Student Label

Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

Step 5: Verify in the Final Dashboard View

